On the Feasibility of Split Learning, Transfer Learning and Federated Learning for Preserving Security in ITS Systems

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***Abstract*— Due to the absence of distinct boundaries, wireless networks are vulnerable to a variety of intrusions. As the number of intruders has increased, the risks on critical infrastructures monitored by networked systems have also increased. Protecting shared information using effective and robust Intrusion Detection Systems (IDSs) remains a critical issue, especially with the growing implementation of vehicular networks. Building an IDS that detects threats efficiently with maximum accuracy and detection is a challenging undertaking. Machine Learning (ML) mechanisms have been successfully adopted in IDSs to detect a variety of network intruders. Split learning is considered one of the main developments in creating efficient ML approaches. In utilizing the Split Learning approach, an IDS is successful in performing at higher accuracy, and detection rate as well as a higher classification performance (Precision, Recall). In this work, a Split Learning-based IDS (*SplitLearn*) for Intelligent Transportation System (ITS) infrastructures has been proposed to address the potential security concerns. The proposed model has been evaluated and compared against other models (i.e., Federated Learning (*FedLearn*) and Transfer Learning (*TransLearn*)-based solutions). With the highest accuracy and detection rates, the proposed model (*SplitLearn*) outperforms *FedLearn* and *TransLearn* by 2 to 5 % respectively. We also see a decrease in power consumption when utilizing *SplitLearn* versus *FedLearn*.**

***Index Terms*— Federated learning, split learning, transfer learning, Internet of Things (IoT).**

I. Introduction

MALICIOUS attacks have become more complex and frequent in recent years as IoT systems have grown larger and encompass different areas of industry. Within the IoT systems themselves, the addition of various types of devices being monitored and used on the overall IoT system infrastructure has also grown and become more diverse [1]. Sensors for monitoring systems to large devices are all part of the Internet of Things [2], [3]. We have also seen a growth in intelligent transport systems (ITS) usage through the utilization of vehicular network architecture. In the healthcare space, we have seen an increase in IoT medical devices. With all these components now becoming prominent, there is a need for proper monitoring. The large array of devices that are controlled, monitored, or used produces networks that are full of security breaches.

On the other hand, wireless, Bluetooth, or LAN are all distributed networks that give way to connecting many IoT devices [4]. Malicious users will always take advantage of any weak point in the network, seeing as IoT devices can be connected to different types of network infrastructure they become the prime access point for attack. Specifically in health care, an IoT edge device that is connected both to Bluetooth and the main LAN healthcare infrastructure becomes an easy gateway to infiltrate. Fast production and distribution in order to keep up with the ever-changing IoT technology market makes it difficult for companies to test and mitigate several security issues. In addition, with the increase of different outlets for IoT device production, standards and protocols will differ on devices connected to the same network. The overall challenges that are faced vary as we become more accustomed to IoT devices in our daily lives. Considering that IoT devices run on different communication protocols, can an implemented IDS detect types of attacks for all types of protocols? With the increasing influx of data that the IoT health care network deals with on a daily basis, can the IDS detect malicious attacks in an efficient time frame and deploy a viable solution?

IDS as a whole is a structure that manages many different subsystems such as healthcare, transportation, home, or business systems. It also can be a combination of systems that have been listed. Given this, many researchers develop IDS that are application-specific for the components in the network they are trying to protect. The contributions in this paper are summarized as follows. First, we focus on creating a more secure and efficient ITS, across sectors of data acquisition, aggregation, and analysis, with this comes many obstacles. Section II takes into account the previous work that was completed through the implementation of a healthcare system, it has now been expanded into a generalized framework for an ITS infrastructure.

It has been predicted that by 2022, there will be a great increase in the number of devices that are embedded in a given ITS infrastructure. Due to the large amount of data

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and the generalized implementation, we have expanded our model to not only utilize federated learning but also looked into other machine learning models such as split and transfer learning. These models will generalize well on unseen data and will provide a more secure access flow for the transmitted information.

Federated and split learning are utilized hand in hand, dependent on where you are on the system's network and architecture. Federated in many cases has a faster performance than Split due to its use of a parallel client-side model generation. Split's architecture of separating server and client is the better option when it comes to privacy [5]. Both are popular due to their privacy-preserving capabilities and are used to that advantage. Utilizing both models provide the ability to implement it on a wide variety of intelligent system structures. With real-time data becoming more prevalent in intelligent systems, IDS models need to ensure the effective performance handling. Transfer learning has been developed to take advantage of this fact. Transfer learning models are a way to reuse the structure for multiple tasks, for example when fine-tuning is one example of the reuse of the model for multiple tasks in the system [6].

The remainder of the paper is organized as follows: Section II covers the prior work that has been done to answer the aforementioned issues through the use of a variety of approaches. Section III discusses the main components of the proposed model and how it would be built to address the problems stated. Section IV discusses the performance evaluation of the proposed model. Section V lists our findings that were produced from the proposed model in addition to some future work.

II. Related Work and Motivation

ITS is an umbrella term for many kinds of systems, there are many options to make sure that the implementation of the system has the maximum amount of impact in terms of information delivery and analysis. Recently the decision to create an efficient ITS infrastructure is impacted by the growing challenge of preserving the security and privacy of system users. Specifically with ITS becoming a more prominent structure in a future that complements fully automated vehicles, security becomes of the utmost importance. Attack types in terms of IoV can be classified in one of two groups, inter-vehicle and intra-vehicle attacks. The former information is exchanged between vehicles in the network, this is a cause for a concern when looking at specific attacks such as Sybil or flooding attacks in the vehicular network. The latter looks into the communication that is being exchanged within a specific vehicle such as GPS information and other sensor information, if this information was altered it can cause not only problems within that specific vehicle but also lead to inter-vehicle attacks [7]. The structure and architecture of an ITS appears to emulate the architecture of a healthcare IoT system with similar sources of faults where a network attack can occur. With that in mind, we explore previous research in terms of the privacy and security of IoV networks.

Below we go through some of the past research in the field of IoT security, as well as how it was applied to various data sets and networking structures. In general, machine learning tools have been at the forefront in terms of their application when it comes to ITS and vehicular networks. Supervised learning to combat anomaly detection, unsupervised learning for congestion control and reinforcement learning for routing and resource management [8]. We have to look for solutions that not only improve security and privacy but also can deal with the ever-changing environment that is IoV, keeping in mind that these models have the ability to be implemented in other network structures and scenarios.

Authors in [9] propose an intelligent multi-hop broadcast protocol which helped them maintain a more flexible and dynamic structure to combat the ever-changing network environment. The proposed protocol employs a fuzzy logic-based approach with the addition of transfer learning to improve speed when new vehicles entering the network change the structure. They were successful in decreasing end-to-end delay and increasing packet dissemination.

In [10], authors utilized a Hybrid Deep Transfer Learning Method to maximize capacity on a 5G vehicular network. Researchers concluded that the method proposed showed an increase in traffic throughput when a new object changed or entered the network.

In order to improve the overall machine learning method in terms of forecasting private data leakage in industrial IoTs (IIoTs), the authors of [11] use federated learning. Furthermore, a blockchain data sharing infrastructure is developed to improve data sharing privacy. The combination of these three technologies is shown to be a viable solution for preventing data leakages in distributed IIoT systems using two unstructured short text data sets that simulate the IIoT.

The application of semi-supervised deep reinforcement learning to improve the environment for smart cities and indoor localisation is explored in [12]. When they used a mix of labeled and unlabeled data, they found a 27% improvement in accuracy in indoor localisation. In terms of IoT and smart cities, the findings demonstrate that reinforcement learning may be used in a variety of ways.

According to [13], a federated cloud network model applied to the IoT gateway enables transparent communication among IoT devices. The authors use the network function virtualization idea as an additional structure to provide safe services across all devices in the IoT network. A federated security architecture has also been proposed in [14] to enable a more secure interchange of IoT services. Both concepts have been proposed in order to start a discussion and develop a future paradigm.

Authors in [15] looked into federated learning reliability in mobile networks. This correlation study between mobile and IoT networks is of great importance due to their examination of investigating an effective way to secure communication on decentralized network infrastructure. An underlying reputation component was created through the implementation of a block-chain structure scheme. This scheme was utilized as an added layer so as to not deceive the federated learning model layer. This question for this study remains in the dynamic optimization of malicious attackers when the amount of workers increases in the network. For authors in [16] they

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apply a neural network federated learning structure towards securing collaborator communication in terms of accessing, collecting and analyzing shared data collections. To test their structure, a very small data set is used with an accuracy output of 54%. This gives way to more open research to create a more reliable prediction accuracy.

The research presented in this paper focuses on implementing a generalized federated reinforcement learning model for ITS. Although, several researchers are also focusing on healthcare and federated learning. For example, Xu and Wang take on the challenge of reviewing how federated learning can be highly beneficial when utilized to safeguard and assess health care informatics [17]. Our research is presented to bridge the divide between IoT healthcare devices, healthcare informatics, and detection of intrusion that could be a massive interference in these areas. Another area that has recently been revitalized due to the introduction of transfer and split learning is the bio-informatics sphere. Authors in [18] proposed to take advantage of transfer learning techniques to detect and diagnose Myocardial infarction (MI) in urban healthcare in support of reducing over-fitting in a convolution neural network detection model. The accuracy produced was between 98 to 99.5 % giving a new life to MI detection. Similarly, in [19] authors utilized multi- source adversarial transfer learning framework to produce better quality information coming from multiple sources of data to predict glucose levels in diabetics. This method gives a more general feature representation and improves the efficiency of detection of an otherwise unseen target.

From the analysis of previous research, we have found that a federated learning structure and transfer learning are the perfect application to strengthen the security of the ever-changing vehicular network systems. In addition, we have seen that this type of application will not only benefit ITS but also could be used in other systems such as IIoT, bio-informatic healthcare structures, and smart cities. Federated, Split and transfer learning algorithms ensures device security within the network with added security against any future incoming data and communication from devices or other networks.

All previous research in this domain handle their security needs by mixing a reinforcement learning model with a blockchain infrastructure. We propose to take this a step further and compose a federated reinforcement learning structure that will not only combat attacks on the IoV network but can be generalized to be utilized in other forms of IoT architectures.

III. Proposed Model

A. Transfer Learning ($TransLearn$)

The reuse of a previously learned model on a new problem is known as transfer learning. It is particularly popular in deep learning due to its ability to train deep neural networks with a small amount of data. This is particularly valuable in the field of data science, as most real-world situations do not require millions of labeled data points to train complicated models. Aiming to use what is learned in one task to boost generalization in another task. Through transfer learning, the main concept is to apply what a model has learned from a task with a large amount of labeled training data to a new task with less data. Rather than starting from zero, $TransLearn$ starts with patterns gained by completing a related task. Transferring as much knowledge as possible from the previous task to learn the new task. Transfer learning provides a number of advantages, the most important of which are reduced training time, improved performance, and the absence of a large amount of data.

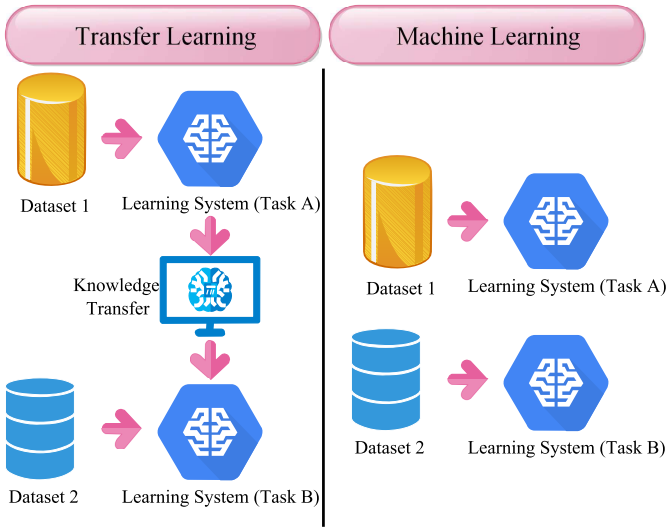

Fig. 1. Transfer learning vs. traditional machine learning representation.

Considering IoTs and with the variety of its devices (range from tiny devices to huge devices) and the variety of data size, transfer learning could be applied since some devices might not have enough labeled training data to start from scratch and train the network, some devices might have a large amount of data while other devices have extra small amounts of data as well as some tasks might be almost the same.

Fig. 1 shows the simplest comparison between the traditional machine learning with the transfer learning.

B. Federated Learning ($FedLearn$)

$FedLearn$ is a collaborative Machine Learning (ML) system in which users maintain their locally acquired data sets on their devices and train their own models on those devices. The global model is then constructed by averaging the locally trained models [20], [21]. The federated learning mechanism is represented in Fig. 2. Client devices do not share their local training and testing data, which could contain private measurements and information. In return client devices, train their local models and only share the parameters of those models with the server. From the perspective of an ITS infrastructure, utilizing the $FedLearn$ collaborative model will aid in scalability and user privacy as more and more IoT devices are introduced to the system.

$FedLearn$-Averaging algorithm ($FedLearn_{avg}$) is contingent on the training being coordinated within the centralized server that creates the shared global model. Over each round of data collection, $FedLearn_{avg}$ works with three parameters [22]: 1) The fraction of clients doing calculations, 2) The number of training activities on local datasets, and 3) The local

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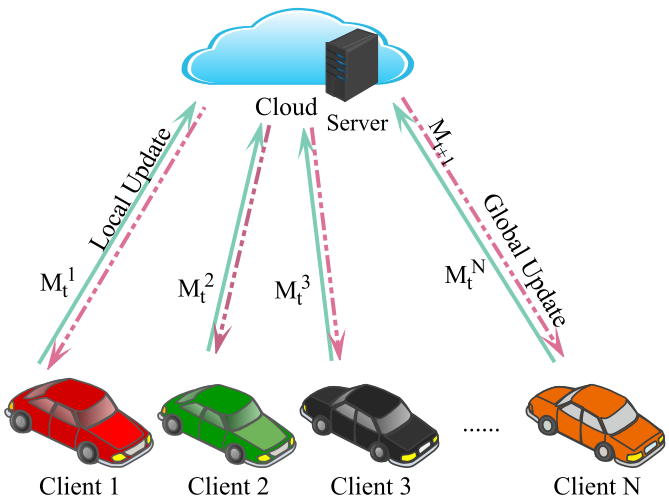

Fig. 2. Federated learning representation.

mini-batch size as shown in Algorithm 1. These parameters aid in the distribution of gradient descent load among clients. Subsequently, the models are averaged on the server and make a dynamic change. The server does this by aggregating the information received by the client and broadcasting the new models and parameters for the following round of data collection.

Algorithm 1 Federated Learning Procedure

procedure FEDERATED LEARNING(Server side)
W_0: Server initialized model
  for each iteration **do**
Perform $|C_t| = \mathbb{C}.c \geq 1$
    for each client $c \in |C_t|$ **do**
Update the user w_{t+1}^u
    $w_{t+1} = \sum_{c \in \mathbb{C}_{\approx}} \frac{n_c}{n_\alpha} w_{t+1}^c$
procedure FEDERATED LEARNING(User side)
Receive w_t from the server
Initialize $w_{t,0}^c = w_t$
  for h= 0,1,2,... **do**
Select a sample β from D_{in} ▷ D_{in} = user dataset **Update** $w_{t,h+1}^c = w_{t,h}^c - (m_{t,h}^c, \beta)$
  Set $w_{t+1}^c = w_{t,H_t}^c$
Send w_{t+1}^c back to the server

C. Split Learning (*SplitLearn*)

SplitLearn is a newly developed technique that allows for participating entities to train machine learning models without sharing any raw data. Split learning is a distributed deep learning methodology in which a CNN is split into two portions, one for the client and the other for the server.

Fig. 3 represents the split learning mechanism. *SplitLearn* considers both forward propagation and backpropagation in which they carried out in the following manners: A client trains the network with raw data up to a specific layer of the network, known as the *cut layer*, as shown in Fig. 3, and then provides the activations of the cut layer, also known as *Smashed data*, to the server. After that, the server uses the

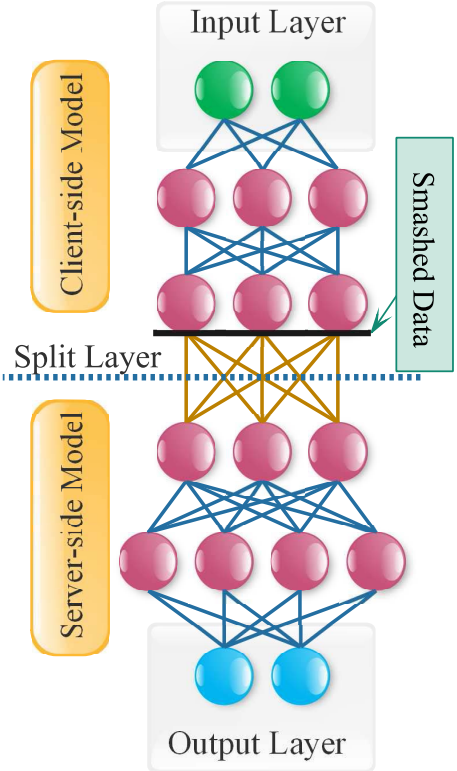

Fig. 3. Split learning representation.

crushed data it received from the client to train the subsequent layers. Thus, a single forward propagation is now complete. The server then does backpropagation up to the cut layer and delivers the crushed data gradients to the client. The client uses the gradients to perform backpropagation on the remaining network (i.e., up to the first layer of the network). Within this, a single pass of backpropagation between a client and the server is now complete. This back-and-forward process continues until the network has been trained with all of the available clients and has reached convergence.

Table I Represents the notations introduced in the equations, system model and algorithms.

Algorithm 2 summarizes the split learning (*SplitLearn*) method.

D. Split Learning (*SplitLearn*)-Based Security Preserving Solution

In this subsection, we present the general structure for the adopted *SplitLearn* in Intelligent Transportation System (ITS) to build an Intrusion Detection System (IDS) model. The *SplitLearn* technique has been applied to classify the traffic and predict whether traffic is an attack or normal. As presented in Fig. 4, the proposed model allows the participants to train machine learning models without sharing any raw data, such as different data stores collecting local data, which are then trained and sent over as a compressed structure to the AI cloud again represented in Fig. 4. This first component describes the client-side of the Model. In *SplitLearn*, the architectural setups are carried out by a trusted third party with direct access to the server. In this case,

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TABLE I
THE MOST USED NOTATIONS IN THE PAPER

Notation	Notation Description
i	Round
s, S	S users (clients) in total ($s \in \{1, 2, \ldots, S\}$)
M	The model
$M^U_{s,t}$	Client-side model of user s at t
M^R_t	Server-side model at t
B	Set of batches of data
b	Batch
w_0	Server' global model
c_t	Set of clients
w_t	Current global model
w^C_t	Updated model for all clients
β	Size of subsets splitted by clients
$Iter$	SGD iterations
w^C_{t+1}	Individually trained models by clients
w_{t+1}	Recently renewed global model
C	Clients fraction
β	Size of subsets (batches)
$Iter$	Iteration number
Γ	Learning rate
λ	Learning rate decay
FP	False Positive Occurrences
TP	True Positive Occurrences
FN	False Negative Occurrences
TN	True Negative Occurrences

Algorithm 2 Split Learning ($SplitLearn$)

procedure
Server Side at round i $\geq$ 0:
  for request r from user $s \in S_t$ **do**
$(A_{s,t}, Y_s) \leftarrow UserUpdate(M^U_{s,t})$. ▷ $A_{s,t}$ and Y_s from user s
Track the forward propagation on M^R_t ▷ M^R_t is the server side of M
Calculate loss with Y_s
Track the backpropagation and model updates
Send the updates to the user s for the ***UserBackPropagation***$(dA_{s,t})$
  end
User Side
ValidateUserUpdate $(M^U_{s,t})$
  if User s is first user for training **then**
$(M^U_{s,t}) \leftarrow$ Random initialization
else
$(M^U_{s,t}) \leftarrow$ ***UserBackPropagation***$(dA_{s-1,t-1})$
end
ValidateUserBackPropagation$(dA_{s,t})$:
  for each batch $b \in B$ **do**
    Backprop with $(dA_{s,b,t})$
    Model updates M^R_{t+1}
  end
Send M^R_{t+1} to the next user ready to train with the server

the authorized party chooses the machine learning model and network splitting (how to identify the cut layer). The learning process is synchronized with numerous clients in either a centralized or peer-to-peer approach: 1) centralized mode: a client updates its client-side model before initiating training with the server by obtaining it from a trustworthy third-party server, which keeps the updated client-side model supplied by the previous trained client. 2) peer-to-peer mode: the client updates its client-side model by downloading it directly from the last taught client. Overall, the training is done in a relay-based manner, with the server first training one client and then moving on to the next client in a sequential manner. The model is collaboratively trained by both the clients and the server without exposing one other's portions, the smashed data has been directed from the clients' side to the server. The smashed data gradients have been forwarded from the server toward the clients. The $SplitLearn$ steps are represented in Algorithm 2.

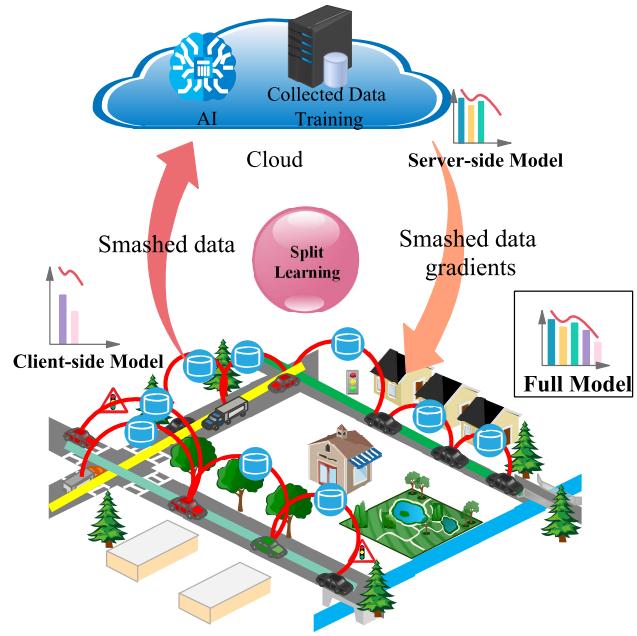

Fig. 4. SplitLearn model.

IV. PERFORMANCE EVALUATION

A. Description of the Dataset

For the proposed model evaluation, we adopted the Canadian Institute of Cybersecurity Intrusion Detection System (CICIDS2017) dataset that has $\approx$ three million samples with fifteen various classes [23]. In order to minimize the number of classes, they merged the few minority attack classes with similar characteristics and behavior, resulting in reducing the number of classes to seven (1 normal + 6 attack). We split the data in 75% for training and 25% for testing). The dataset has eight files that summarize five days of normal and attacks data of the Canadian Institute of Cybersecurity. The data features include Flow ID that contains the Source IP/port and the Destination IP/port, Source and destination IPs, Timestamp that refers to the timing and the Label. The used protocol, Packet Length Min/Max/Mean and Flags are also features. The instances (attacks/normal) are summarized below:

- **DDoS**, When an intruder utilizes many machines to attack a single machine, this is known as a multi-machine attack.
- **DoS Hulk**, A DoS attack on a web server is carried out using the hulk software tools.
- **DoS slow loris** In which the attacker uses slow loris tools to launch a DoS attack.

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TABLE II
CICIDS2017 DATASET ATTACKS/NORMAL OCCURRENCES

Traffic Types (Normal/Attack)	Number of entities
DDoS	41835
DoS Hulk	231072
DoS slow loris	5796
DoS GoldenEye	10293
DoS SlowHTTPTest	5499
Web Attack – XSS	652
Web Attack – Brute Force	1507
Web Attack – SQL Injection	21
Infiltration	36
Bot	1966
PortScan	158930
Heartbleed	11
SSH-Patator	5897
FTP-Patator	7938
Benign	2359087

- **DoS GoldenEye** when the attacker uses GoldenEye tools to launch a DoS attack.
- **DoS SlowHTTPTest**, The attacker sends HTTP requests to a web server in small chunks.
- **Web Attack – XSS** that is when an attacker uses a web application to transmit malicious code to an end device, this is known as a cross-site scripting (XSS) attack
- **Web Attack – Brute Force**, The machine password is cracked using a search-based approach that anticipates all possible combinations.
- **Web Attack – SQL Injection**, An injection attack that allows the execution of malicious SQL statements.
- **Infiltration**, The attacker has already gained access to a system after successfully compromising it.
- **Bot**, Trojans are used to breach the security of machines that have been targeted.
- **PortScan**, By routing packets with different destinations' ports, the attacker gathers information about the targeted computer, such as the running operating system.
- **Heartbleed**, The Heartbleed issue allows a person to get access/read-protected system memory.
- **SSH-Patator**, Using the SSH-Patator, the attacker uses BFA to guess the SSH password.
- **FTP-Patator**, Using the FTP-Patator, the attacker uses BFA to guess the FTP password.
- **Benign**, refers to normal traffic occurrences.

B. Simulation Settings

Various tests were performed to examine different aspects of the system, such as, accuracy and detection rates, ROC analysis, and QoE. To simulate the effectiveness of the proposed solution, we have considered a number of experimental environments. The algorithms training and testing was implemented using Python and scikit-learn ML library. The network simulation, connectivity and mobility were tested using NS3 simulator and its supported solutions. We run the clients and server on different high-performance computing nodes.

In training the model, each client trains the network up to the split layer and delivers the weights to the server which then trains it for the remaining layers (where the forward propagation is now complete). The server then creates the gradients for the last layer and propagates the error all the way back to the split layer. After that, the gradient is sent to the client which completes the rest of the backpropagation. This process is repeated until the entire network has been learnt.

TABLE III
SIMULATION PARAMETERS

Simulation Parameter	Value
Training and testing implementation	Python, Scikit-learn ML
Number of nodes	20
Simulator	NS3
Area dimensions	$100m \times 100m$
Communication range	100m
Simulation time	800s
Packet size	250 bytes
Tested transfer learning technique	DenseNet
Observation window	200 global epoch
Batch size	1024

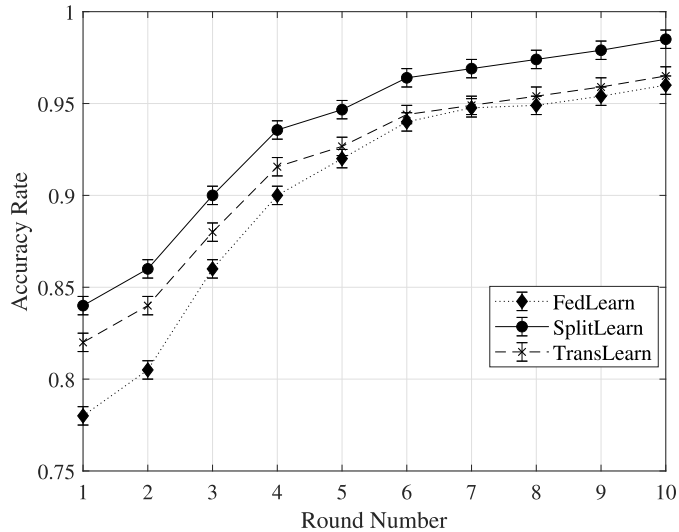

Fig. 5. Accuracy rates comparison of different learning techniques (Federated, Split and Transfer Learning).

Table III summarizes some of the simulation settings.

C. Results Analysis

Several results were achieved through experimentation and are listed and analyzed below.

1) Accuracy and Detection Rate: The Accuracy rate relates to the truly categorized patterns to the false occurrences [24]. The following equations (1, 2) highlights the adopted formula for the accuracy rate calculations. Where True Occurrences refer to the truly detected patterns (ex. TP and TN), TP refers to the True Positive, TN to the True Negative, FP to the False Positive, and FN to the False Negative.

$$Accuracy = \frac{TrueOccurrences}{All} \tag{1}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

Detection rate relates to the occurrences of malicious instances. It considers the TP occurrences ratio as shown in Equation (3) [24].

$$Detection = \frac{TP}{TP + FP} \tag{3}$$

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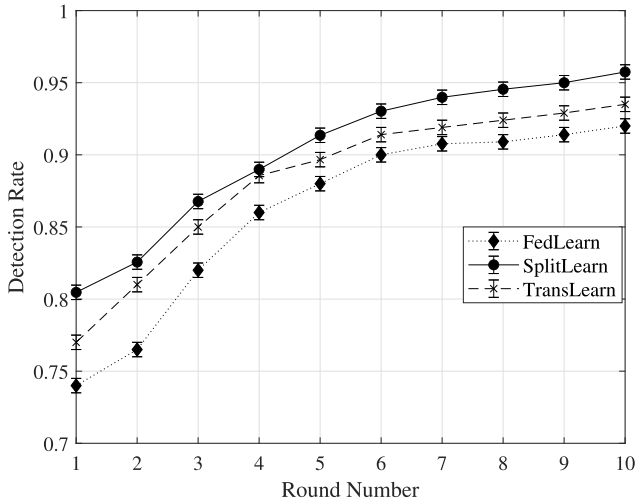

Fig. 6. Detection rates comparison of different learning techniques (Federated, Split and Transfer Learning).

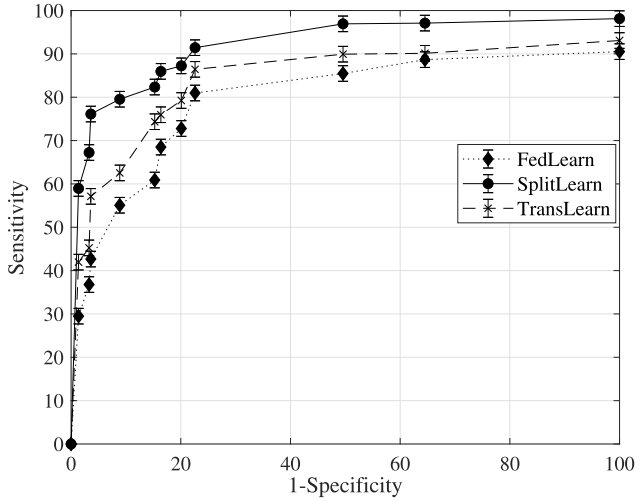

Fig. 7. ROC comparison of different learning techniques (Federated, Split and Transfer Learning).

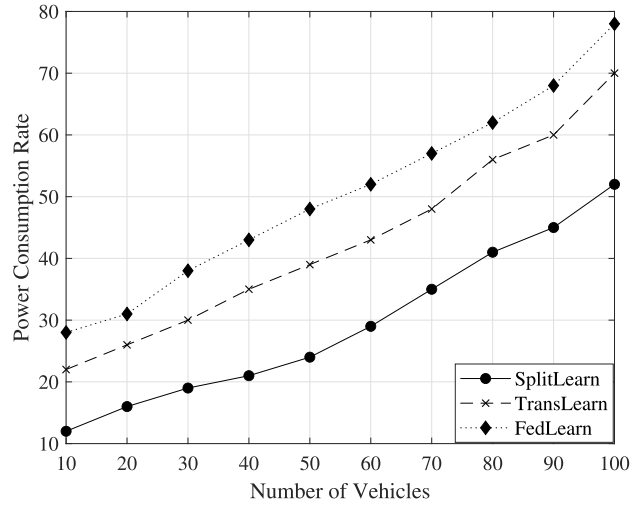

Fig. 8. Power consumption rates comparison of different learning techniques (Federated, Split, and Transfer Learning).

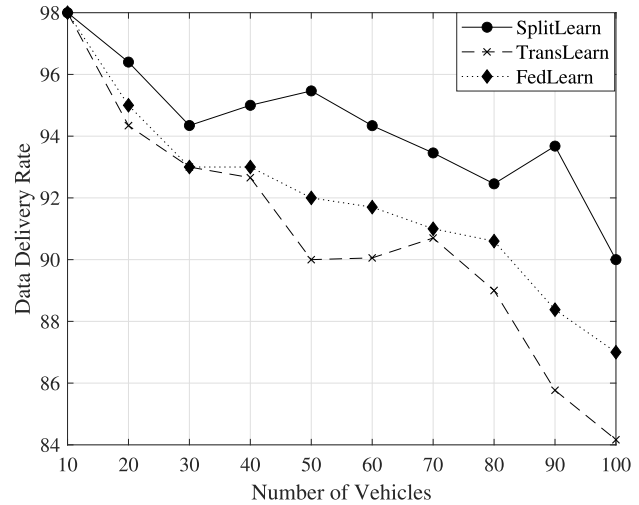

Fig. 9. Packets delivery rates comparison of different learning techniques (Federated, Split, and Transfer Learning).

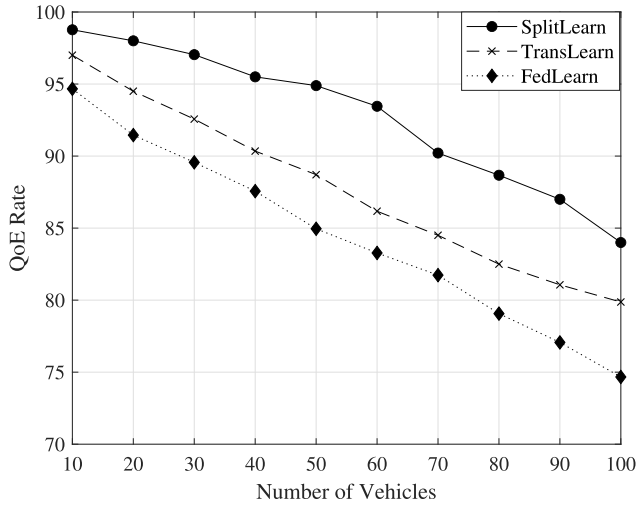

Fig. 10. QoE rates comparison of different learning techniques (Federated, Split, and Transfer Learning).

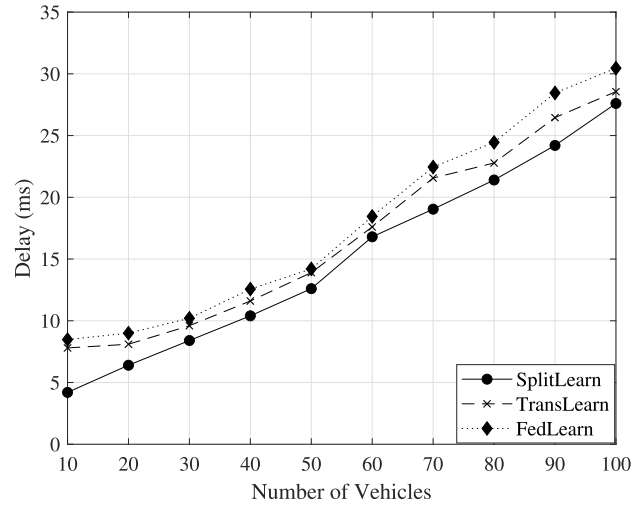

Fig. 11. Delay (ms) comparison of different learning techniques (Federated, Split, and Transfer Learning).

Fig. 5 represents the accuracy rate ($Accuracy$) of the proposed $SplitLearn$ model compared to $TransLearn$ (adopting transfer learning) and $FedLearn$ (adopting federated learning). The graph shows that the proposed $SplitLearn$ outperforms the other models with the highest $Accuracy$ (≈ 0.985) followed by $TransLearn$

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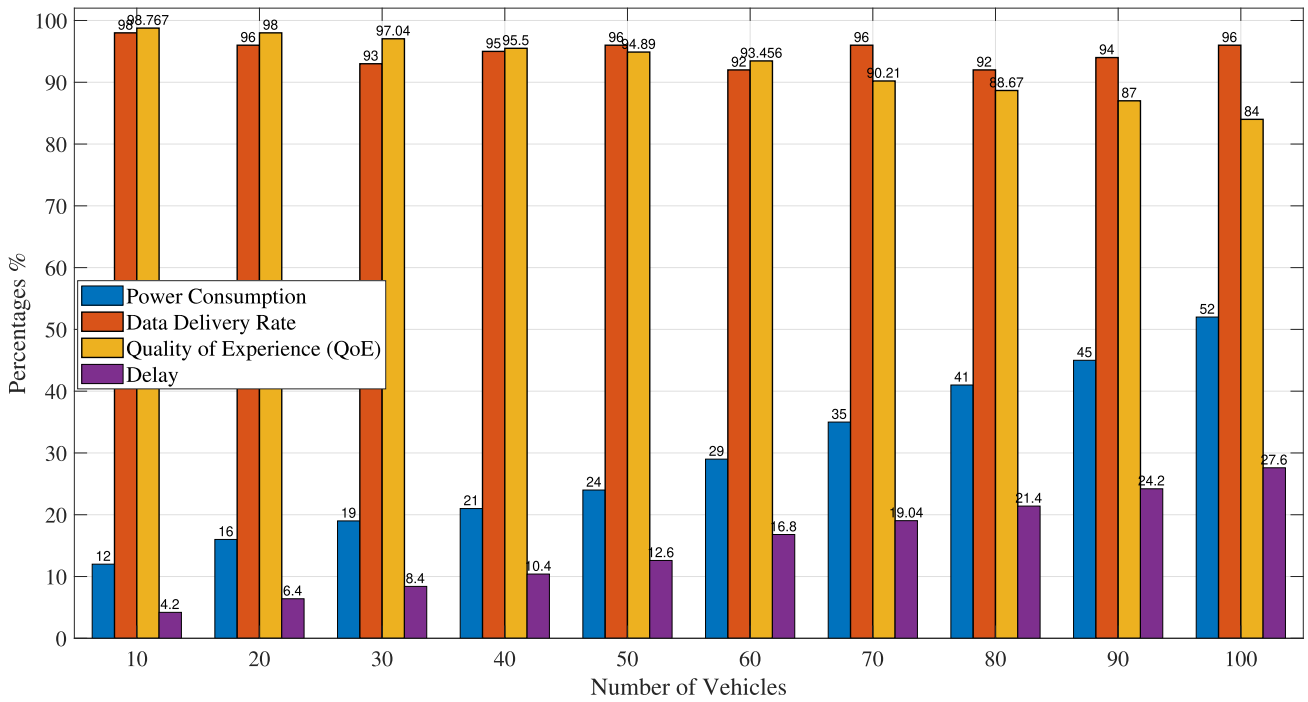

Fig. 12. *SplitLearn* performance evaluation in terms of QoE, Power consumption, Delay, and the packets delivery ratios.

with an $Accuracy$ of $\approx$ 0.968 and $SplitLearn$ with ≈ 0.96.

Fig. 6 illustrates the detection rate $Detection$ of Split Learning ($SplitLearn$), Federated Learning ($FedLearn$), and Transfer Learning ($TransLearn$)-based solutions. When $SplitLearn$ is compared to others (as shown in the figure), it achieves the highest $Detection$ of ($\approx$ 0.96) followed by $TransLearn$ with ≈ 0.94 and $SplitLearn$ with ≈ 0.92.

2) ROC Curve: The Receiver Operating Characteristic (ROC) curve displays the ratio between the $Sensitivity$ (TP) and the FP ($1 - Specificity$). The Sensitivity-specificity is represented by the area under the curve such as the largest the area the best the performance is.

$SplitLearn$, $FedLearn$ and TranLearn curves' are displayed in Fig. 7. $SplitLearn$ outperforms the other models according to its better TP.

3) Power Consumption Rate: Fig. 8 depicts the estimated power consumption at the proposed model. Adopting split learning reduces the overall energy consumption by $\approx$ 20% against the other solutions. This is due to the fact that the learning process has been split between the server and the individual vehicles instead of learning the models on the vehicles.

4) Packets Delivery Ratio: Fig. 9 represents the ratios of the successfully delivered data at the proposed model ($SplitLearn$) against the other models with respect to the number of vehicles. The ratios are all high but it is obvious that adopting $SplitLearn$ has outperformed the other models with $\approx 8\%$.

5) Quality of Experience (QoE): For the QoE evaluation, we have considered three metrics; the delay, energy consumption and the packet dropped ratio. The delay metric refers to the end-to-end latency needed for a vehicle to receive the data. The energy consumption metric returns the maximum consumed energy on the vehicle, and the packet dropped ratio refers to the unsuccessfully delivered packets to the vehicles.

Fig. 10 shows the QoE rate at the proposed model ($SplitLearn$) against the other two models ($TransLearn$ and $FedLearn$). Adopting $SplitLearn$ upgrades the QoE by $\approx$ 10% against the other solutions. This is due to the fact that the $SplitLearn$ performs with less energy consumption, less latency, less packet dropped rates and better packet delivery rates.

6) Delay Analysis: Fig. 11 represents the ratios of the delay in the proposed model ($SplitLearn$) against the other models with respect to the number of vehicles. The delay refers to the end-to-end latency in delivering the data to the vehicles. It is clear that adopting $SplitLearn$ has outperformed the other models with $\approx 5ms - 10ms$.

Fig. 12 represents a comprehensive view of the estimated power consumption, QoE and the overall packet delivery rates adopting the $SplitLearn$ model.

V. Conclusion

In cruric infrastructures, security has a major role in the credibility of any system. In this paper, a secure IDS model has been proposed by applying split learning.

Within simulations, we have compared our proposed $SplitLearn$ with other solutions namely $FedLearn$ and $TransLearn$ and confirmed that the $SplitLearn$ solution outperforms the other solutions in terms of accuracy, detection rates, area under the curve (ROC), power consumption, packets delivery ratio, Quality of Experience (QoE) and delay analysis respectively, in the presence of malicious actions in the experienced ITS. The proposed model ($SplitLearn$) outperforms $FedLearn$ and $TransLearn$ by 2 to 5 % respectively.

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We also see a decrease in power consumption when utilizing $SplitLearn$ versus $FedLearn$.

Extension of the proposed solution to take into account various networks with heterogeneous device types. Furthermore, evaluating the influence of various emerging learning models, such as deep learning, on such crucial infrastructure with different datasets.

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